SIGMIDI Foot Controller v0.1 — KiCad schematic

`sigmidi-foot-controller.kicad_pro` is an editable KiCad 10 assembly project. The single A0 landscape sheet contains **57 physical components, 49 named nets and 178 connected terminals**. Three additional PWR_FLAG symbols declare externally supplied power and the diode-OR MIDI_5V rail; they are not purchased parts. All connections use continuous wires. Net labels name those wires, and dots identify junctions. Purchased boards stop at their external pins.

- Browser preview
- Vector schematic and PDF
- Electrical drawing and connection schedule
- `SIGMIDI.kicad_sym` and the project-relative `sym-lib-table` provide the custom external-module symbols.

Physical pin references

U1 is Arduino Nano ESP32 ABX00083, component-side top view with USB-C up. Its **JP1 is on the left, pads 1–15 top to bottom; JP2 is on the right, pads 15–1 top to bottom**. Pad IDs come from Arduino schematic V0.3; physical positions and D/A labels come from its official top-view pinout. The datasheet's functional-table row numbering is not used as JP2 pad numbering. VBUS is named VUSB in the pinout, and RST is RESET. Every printed header label remains visible, including unused pins; connected pins also show the ESP32 GPIO. The firmware uses GPIO numbers independently of Arduino's pin-numbering preference.

The Waveshare 24382 external P2 connector retains **1 BL, 2 RST, 3 D/C, 4 CS, 5 SCL/CLK, 6 SDA/DIN, 7 GND, 8 VCC**. SCL/SDA are SPI signals. The display's internal FPC and circuitry are omitted.

Pololu 2535 LEDs use the actual package lead order: lens left, leads right, flat side down; top to bottom DIN, 5V, GND, DOUT. DIN/DOUT/+5V/GND are functional terminal identifiers, not invented package pad numbers. LED1 receives D7/GPIO10 directly; there are no added LED-chain parts. The final DOUT is marked NC.

J2 is ACJS-MHDC: manufacturer TOP is pedal 1 (10 sleeve, 11 ring, 12 tip); BOTTOM is pedal 2 (4 sleeve, 5 ring, 6 tip). Normal contacts are NC and pad 13 is mechanical. SW11 uses functional A/B/C/SW1/SW2 names; confirm the selected encoder terminal layout before soldering. The passive encoder uses internal GPIO pull-ups without external R/C.

Nano VIN receives the barrel sleeve's +9V. The Pololu converter's 5V output powers the optional LEDs and D1. USB VBUS feeds R50 100k / R51 150k, with their junction at A4/GPIO11, and D2. D1/D2 1N5819 diode-OR +5V and USB VBUS into MIDI_5V. ADC inputs are A0/GPIO1, A1/GPIO2 and A3/GPIO4; pedal bias outputs are D2/GPIO5 and D3/GPIO6.

MIDI OUT uses D0/GPIO44 (UART1 TX, MIDI_OUT_TX) with R52 10k pull-up to 3V3. U4 74HCT125 runs from MIDI_5V with C6 100nF across pins 14/7. Gates 1 and 2 are used: inputs 2 and 5 on MIDI_OUT_TX, enables 1 and 4 to GND; unused inputs 9/12 and enables 10/13 go to GND and outputs 8/11 are open. J3 (MIDI OUT A) and J4 (MIDI OUT B) are 3.5 mm TRS Type A sockets: tip is the sink through R54/R56 220R from U4 outputs 3/6, ring is the source through R53/R55 220R from MIDI_5V, and sleeve is GND. U4 is drawn as one functional unit with DIP/SOIC-14 package pin numbers.

Validation and scope

The saved schematic is checked with direct KiCad ERC and XML exports. `verification/checks.json` records the latest findings, tool versions and source/output hashes. `verification/pin-locations.json` records queried symbol coordinates. `verification/physical-pin-map.json` ties accepted terminal IDs and labels to those positions. Run `scripts/verify_kicad_netlist.py` to compare every exported net against the electrical drawing and verify physical Nano/LCD pin order independently of placement requests. Package pins absent from the electrical pin map, such as U4 outputs 8 and 11, must export as single-pin nets.

No PCB footprints are assigned. The empty `.kicad_pcb` is a project placeholder; no routed board, manufacturable PCB or hardware qualification is claimed. Standard resistor, capacitor and push-switch symbols

are used for discrete parts. Custom symbols describe purchased modules and functional terminals, and give the 74HCT125, 1N5819 diodes and TRS sockets the terminal IDs used by the electrical pin map. Component fit, noise, power behaviour and direct LED data levels require bench verification.

Edit the schematic in KiCad or through Konnect MCP. Export the netlist, SVG and PDF with `kicad-cli sch export netlist --format kicadxml`, `kicad-cli sch export svg` and `kicad-cli sch export pdf -o output/pdf/sigmidi-kicad-schematic.pdf`, render the PNG from the SVG at 4096 px width, then run `scripts/preview_kicad.py` to publish the verified preview.